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A primer on importance sampling and particle filters

Víctor Elvira
School of Mathematics
University of Edinburgh
(victor.elvira@ed.ac.uk)

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Outline

State-space models and Bayesian filtering

Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

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Particle filtering from the MIS and AIS perspectives

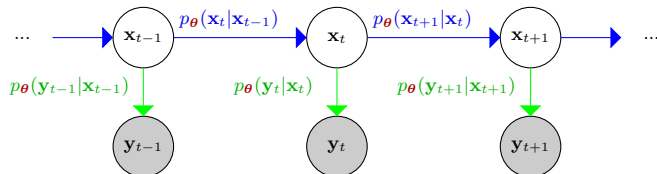
Advanced particle filtering

- PFs in high-dimension spaces

- The choice of N in PFs

1. Modeling: state-space models (SSM)

- Time-series data are collected, $\mathbf{y}_t \in \mathbb{R}^{N_y}$, $t = 1, \dots, T$:
- A SSM models a sequence of hidden states $\mathbf{x}_t \in \mathbb{R}^{N_x}$, $t = 1, \dots, T$.



- Probabilistic notation of a (simple) Markovian SSM:
 - state model $\rightarrow p_{\theta}(\mathbf{x}_t | \mathbf{x}_{t-1}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}, \theta)$
 - observation model $\rightarrow p_{\theta}(\mathbf{y}_t | \mathbf{x}_t) = p(\mathbf{y}_t | \mathbf{x}_t, \theta)$
 - prior on initial state $\rightarrow p_{\theta}(\mathbf{x}_0) = p(\mathbf{x}_0 | \theta)$

2. Estimation/inference problems

- We sequentially observe data $\mathbf{y}_t$ related to the hidden state $\mathbf{x}_t$.
 - At time t , we have accumulated t observations, $\mathbf{y}_{1:t} \equiv \{\mathbf{y}_1, \dots, \mathbf{y}_t\}$.
- Wish list:
 - prediction of future **observations** and estimation of **states** (with uncertainty quantification)
 - * **Filtering**: $p_{\theta}(\mathbf{x}_t | \mathbf{y}_{1:t})$ and joint $p_{\theta}(\mathbf{x}_{1:t} | \mathbf{y}_{1:t})$
 - * State prediction: $p_{\theta}(\mathbf{x}_{t+\tau} | \mathbf{y}_{1:t})$, $\tau \geq 1$
 - * Observation prediction: $p_{\theta}(\mathbf{y}_{t+\tau} | \mathbf{y}_{1:t})$, $\tau \geq 1$
 - * **Smoothing**: $p_{\theta}(\mathbf{x}_{t-\tau} | \mathbf{y}_{1:t})$, $\tau \geq 1$
 - estimation of **model parameters** (with interpretability)
- Bayesian/probabilistic inference:
 - we compute or approximate pdfs of unknowns when possible (instead of point-wise estimates)



Bayesian filtering

- Bayesian rule for the joint:

$$p(\mathbf{x}_{1:T}|\mathbf{y}_{1:T}) = \frac{p(\mathbf{y}_{1:T}|\mathbf{x}_{1:T})p(\mathbf{x}_{1:T})}{p(\mathbf{y}_{1:T})}$$

- Filtering distribution as a marginal:

$$p(\mathbf{x}_T|\mathbf{y}_{1:T}) = \int p(\mathbf{x}_{1:T}|\mathbf{y}_{1:T})d\mathbf{x}_1d\mathbf{x}_2\dots d\mathbf{x}_{T-1}$$

- Problems:

- **Dimension:** $\mathbf{x}_{1:T} \in \mathbb{R}^{T \cdot d_x}$
- When we receive $\mathbf{y}_t$, we don't want to **reprocess** $\mathbf{y}_{1:t-1}$

Goal: **efficient and sequential** Bayesian inference

Sequential optimal filtering

- **Filtering Problem:**
 - Distribution of $\mathbf{x}_t$ given all the obs. up to time t , $p(\mathbf{x}_t|\mathbf{y}_{1:t})$
 - Recursively from $p(\mathbf{x}_{t-1}|\mathbf{y}_{1:t-1})$ updating with the new $\mathbf{y}_t$

- Optimal filtering:

1. **Prediction** step:

$$p(\mathbf{x}_t|\mathbf{y}_{1:t-1}) = \int p(\mathbf{x}_t|\mathbf{x}_{t-1})p(\mathbf{x}_{t-1}|\mathbf{y}_{1:t-1})d\mathbf{x}_{t-1}$$

2. **Update** step:

$$p(\mathbf{x}_t|\mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{y}_{1:t-1})}{p(\mathbf{y}_t|\mathbf{y}_{1:t-1})}$$

- Interest in integrals of the form: $I(f) = \int f(\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{y}_{1:t})d\mathbf{x}_t$
 - e.g., the mean, $I(f) = \int \mathbf{x}_t p(\mathbf{x}_t|\mathbf{y}_{1:t})d\mathbf{x}_t$
 - Usually the posterior **cannot** be analytically computed!

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Linear-Gaussian state-space model

- Kalman filter and RTS smoother are the close-form solutions for the linear-Gaussian SSM:
 - Hidden state $\rightarrow p(\mathbf{x}_t|\mathbf{x}_{t-1}) \equiv \mathcal{N}(\mathbf{x}_t; \mathbf{A}_t\mathbf{x}_{t-1}, \mathbf{Q}_t)$
 - Observations $\rightarrow p(\mathbf{y}_t|\mathbf{x}_t) \equiv \mathcal{N}(\mathbf{y}_t; \mathbf{H}_t\mathbf{x}_t, \mathbf{R}_t)$

Kalman filter

- Initialize: $\mathbf{m}_0, \mathbf{P}_0$
- For $t = 1, \dots, T$

Predict stage:

$$\begin{aligned}\mathbf{x}_t^- &= \mathbf{A}_t \mathbf{m}_{t-1} \\ \mathbf{P}_t^- &= \mathbf{A}_t \mathbf{P}_{t-1} \mathbf{A}_t^\top + \mathbf{Q}_t\end{aligned}$$

Update stage:

$$\begin{aligned}\mathbf{z}_t &= \mathbf{y}_t - \mathbf{H}_t \mathbf{x}_t^- \\ \mathbf{S}_t &= \mathbf{H}_t \mathbf{P}_t^- \mathbf{H}_t^\top + \mathbf{R}_t \\ \mathbf{K}_t &= \mathbf{P}_t^- \mathbf{H}_t^\top \mathbf{S}_t^{-1} \\ \mathbf{m}_t &= \mathbf{x}_t^- + \mathbf{K}_t \mathbf{z}_t \\ \mathbf{P}_t &= \mathbf{P}_t^- - \mathbf{K}_t \mathbf{S}_t \mathbf{K}_t^\top\end{aligned}$$

RTS smoother

- For $t = T, \dots, 1$

Smoothing stage:

$$\begin{aligned}\mathbf{x}_{t+1}^- &= \mathbf{A}_t \mathbf{m}_t \\ \mathbf{P}_{t+1}^- &= \mathbf{A}_t \mathbf{P}_t \mathbf{A}_t^\top + \mathbf{Q}_t \\ \mathbf{G}_t &= \mathbf{P}_t \mathbf{A}_t^\top (\mathbf{P}_{t+1}^-)^{-1} \\ \mathbf{m}_t^s &= \mathbf{m}_t + \mathbf{G}_t (\mathbf{m}_{t+1}^s - \mathbf{x}_{t+1}^-) \\ \mathbf{P}_t^s &= \mathbf{P}_t + \mathbf{G}_t (\mathbf{P}_{t+1}^s - \mathbf{P}_{t+1}^-) \mathbf{G}_t^\top\end{aligned}$$

- ✓ Filtering distribution: $p(\mathbf{x}_t|\mathbf{y}_{1:t}) = \mathcal{N}(\mathbf{x}_t; \mathbf{m}_t, \mathbf{P}_t)$
- ✓ Smoothing distribution: $p(\mathbf{x}_t|\mathbf{y}_{1:T}) = \mathcal{N}(\mathbf{x}_t; \mathbf{m}_t^s, \mathbf{P}_t^s)$

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Monte Carlo basics

- Goal:¹ approximate the integral

$$I(f) \equiv \mathbb{E}_{\pi(\mathbf{x})}[f(\mathbf{x})] = \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x}$$

- If we can sample $\mathbf{x}^{(n)} \sim \pi(\mathbf{x})$, $n = 1, \dots, N$, then the target can be approximated as

$$\pi(\mathbf{x}) \approx \pi^N(\mathbf{x}) = \frac{1}{N} \sum_{n=1}^N \delta_{\mathbf{x}^{(n)}}(\mathbf{x})$$

and the moment $I(f)$ can be approximated as

$$I(f) \approx \bar{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)})$$

- This is called raw/crude/direct/standard Monte Carlo.

¹C. P. Robert, G. Casella, and G. Casella. *Monte Carlo statistical methods*. Vol. 2. Springer, 1999.

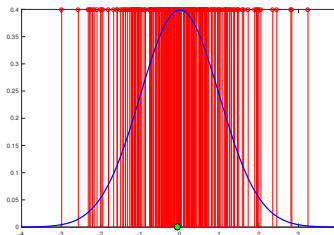
Monte Carlo basics

- We can approximate any integral involving $\pi(\mathbf{x})$: **integral + dirac = sum!**

$$\begin{aligned} \mathbb{E}_{\pi(\mathbf{x})}[f(\mathbf{x})] &= \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x} \approx \int f(\mathbf{x})\pi^N(\mathbf{x})d\mathbf{x} \approx \int f(\mathbf{x}) \left(\frac{1}{N} \sum_{n=1}^N \delta_{\mathbf{x}^{(n)}}(\mathbf{x}) \right) d\mathbf{x} \\ &\approx \frac{1}{N} \sum_{n=1}^N \int f(\mathbf{x}) \delta_{\mathbf{x}^{(n)}}(\mathbf{x}) d\mathbf{x} \\ &= \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}) \end{aligned}$$

- e.g. the mean of the distribution ($h(\mathbf{x}) = \mathbf{x}$) approximated by $N = 200$.

$$\hat{\mathbf{x}} = \mathbb{E}_{\pi(\mathbf{x})}[\mathbf{x}] = \int \mathbf{x}\pi(\mathbf{x})d\mathbf{x} \approx \frac{1}{N} \sum_{n=1}^N \mathbf{x}^{(n)} = -0.0561$$



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Importance sampling (IS) basics

- Unfortunately, we only know how to draw samples from very few distributions.
 - In the general case, we cannot use the raw Monte Carlo.
 - Importance sampling (IS)** is a Monte Carlo method that allows to approximate integrals over complicated distributions.²
- Same problem:

$$I(f) \equiv \mathbb{E}_{\pi(\mathbf{x})}[f(\mathbf{x})] = \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x} = \int f(\mathbf{x})\frac{\pi(\mathbf{x})}{q(\mathbf{x})}q(\mathbf{x})d\mathbf{x}$$

where $q(\mathbf{x})$ is the proposal density where the samples are now drawn

$$\text{Standard MC estimator: } \bar{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}), \quad \mathbf{x}^{(n)} \sim \pi(\mathbf{x})$$

$$\text{Basic IS estimator: } \hat{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}) \frac{\pi(\mathbf{x}^{(n)})}{q(\mathbf{x}^{(n)})}, \quad \mathbf{x}^{(n)} \sim q(\mathbf{x})$$

²V. Elvira and L. Martino. "Advances in importance sampling". In: *arXiv preprint arXiv:2102.05407* (2021).

Importance sampling basics

- Basic IS estimator:

$$\hat{I}_N(f) = \frac{1}{N} \sum_{n=1}^N f(\mathbf{x}^{(n)}) \frac{\pi(\mathbf{x}^{(n)})}{q(\mathbf{x}^{(n)})}, \quad \mathbf{x}^{(n)} \sim q(\mathbf{x})$$

- $W^{(n)} = \frac{\pi(\mathbf{x}^{(n)})}{q(\mathbf{x}^{(n)})}$, $n = 1, \dots, N$, are the **importance weights**
- only constraint: $\pi(\mathbf{x})$ must be evaluated
- Unfortunately, sometimes we have only access to $\gamma(\mathbf{x})$, where $\pi(\mathbf{x}) = \frac{\gamma(\mathbf{x})}{Z}$, Z is the normalizing constant:
 - basic IS estimator is not possible $\Rightarrow$ self-normalized estimator:

$$\tilde{I}_N(f) = \sum_{n=1}^N f(\mathbf{x}^{(n)}) w^{(n)}, \quad \mathbf{x}^{(n)} \sim q(\mathbf{x})$$

where $w^{(n)} = \frac{W^{(n)}}{\sum_{j=1}^N W^{(j)}}$ are the normalized weights ($\sum_{j=1}^N w^{(j)} = 1$).

- * here the weights $W^{(n)}$ can be computed by evaluating $\gamma(\mathbf{x})$ instead.
- Approximation of the targeted distribution

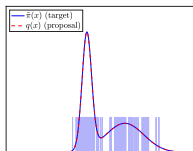
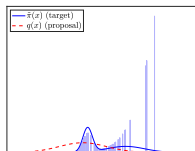
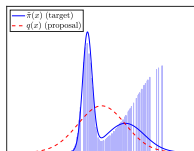
$$\pi(\mathbf{x}) \approx \pi_{\text{IS}}^N(\mathbf{x}) = \sum_{n=1}^N w^{(n)} \delta_{\mathbf{x}^{(n)}}(\mathbf{x})$$

The variance in IS and the need of better proposals

- A **good proposal** $q(\mathbf{x})$ is key for the efficiency of IS.
- Variance of the UIS estimator of $I = \int f(\mathbf{x})\pi(\mathbf{x})d\mathbf{x}$:

$$\text{Var}_{\pi(\mathbf{x})}(\hat{I}) = \frac{1}{N} \int \frac{f^2(\mathbf{x})\pi^2(\mathbf{x})}{q(\mathbf{x})} d\mathbf{x} - \frac{I^2}{N}$$

- optimal UIS proposal: $q(\mathbf{x}) \propto |f(\mathbf{x})|\pi(\mathbf{x})$
- for a generic $f(\mathbf{x})$, $q(\mathbf{x})$ should be as *close* as possible to $\pi(\mathbf{x})$



- Very difficult to find a good $q(\mathbf{x})$ a priori:
 - $\pi(\mathbf{x})$ can be only evaluated (up to a normalizing constant)
 - $\pi(\mathbf{x})$ may be multimodal, skewed, heavy tailed
- A posteriori metric: $\widehat{\text{ESS}} = \frac{1}{\sum_{n=1}^N \bar{w}_n^2}$, although it has known issues³
- Use **multiple proposals** (**MIS**) and **explore** the space (**AIS**).

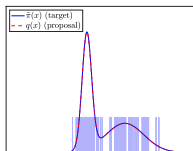
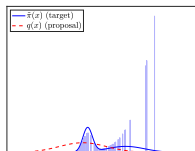
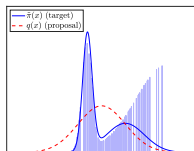
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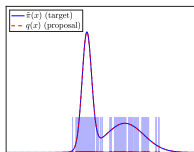
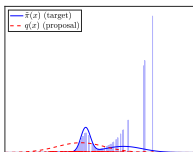
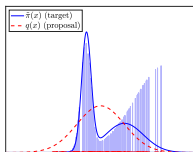
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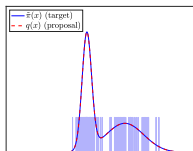
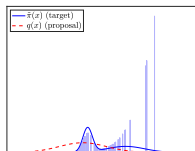
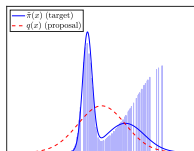
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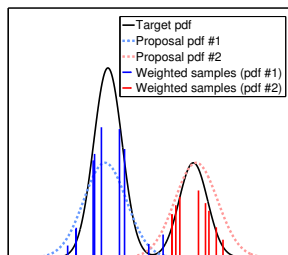
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Multiple Importance Sampling (MIS): Basics

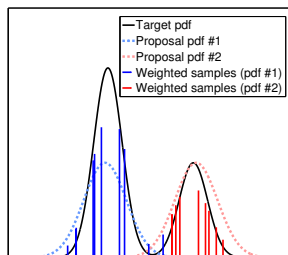
- Set of N available proposal pdfs $\{q_n(\mathbf{x})\}_{n=1}^N$.
 - Example $N = 2$:



- For simplicity, we simulate N samples in total from the set $\{q_n(\mathbf{x})\}_{n=1}^N$, but **how?**
 1. **Sampling:** $\mathbf{x}_n \sim ?$, $n = 1, \dots, N$.
 2. **Weighting:** $W_n = ?$, $n = 1, \dots, N$.
- Most known MIS algorithms focus just in the adaptation (AIS):
 - Implement **sampling** and **weighting** in a different way without much justification.

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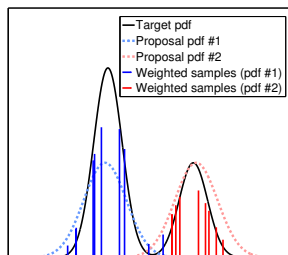
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Multiple Importance Sampling: Interpretations

a) Population Monte Carlo [Cappe04]:

1. **Sampling:** $\mathbf{x}_n \sim q_n(\mathbf{x})$, $n = 1, \dots, N$.
2. **Weighting:** $W_n = \frac{\pi(\mathbf{x}_n)}{q_n(\mathbf{x}_n)}$, $n = 1, \dots, N$.
 - Estimators can be unstable.⁴

b) M-PMC [Douc07a,Cappe08]:

1. **Sampling:** $\mathbf{x}_n \stackrel{i.i.d}{\sim} \psi(\mathbf{x})$, $n = 1, \dots, N$.
2. **Weighting:** $W_n = \frac{\pi(\mathbf{x}_n)}{\psi(\mathbf{x}_n)}$, $n = 1, \dots, N$.
 - Some proposals can be used more than once, others can be not used.

$$\psi(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N q_j(\mathbf{x})$$

c) Deterministic mixture (DM)

1. **Sampling:** $\mathbf{x}_n \sim q_n(\mathbf{x})$, $n = 1, \dots, N$.
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- Why all these sampling/weighting schemes are valid?
- Are some schemes better than others?
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- Novel theoretical framework for MIS $\Rightarrow$ Generalized MIS

⁴O. Cappé, A. Guillin, J.-M. Marin, and C. P. Robert. "Population monte carlo". In: *Journal of Computational and Graphical Statistics* 13.4 (2004), pp: 907–929.

Multiple Importance Sampling: Interpretations

a) Population Monte Carlo [Cappe04]:

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 - Estimators can be unstable.

b) M-PMC [Douc07a, Cappe08]:

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$$\psi(\mathbf{x}) = \frac{1}{N} \sum_{j=1}^N q_j(\mathbf{x})$$

- Some proposals can be used more than once, others can be not used.⁴⁵

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⁵O. Cappé, R. Douc, A. Guillin, J.-M. Marin, and C. P. Robert. "Adaptive importance sampling in general mixture classes". In: *Statistics and Computing* 18 (2008), pp. 447–459. ↻ 🔍 18/76

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⁴E. Veach and L. J. Guibas. "Optimally combining sampling techniques for Monte Carlo rendering". In: *Proceedings of the 22nd annual conference on Computer graphics and interactive techniques*. 1995, pp. 419–428.

⁵A. Owen and Y. Zhou. "Safe and effective importance sampling". In: *Journal of the American Statistical Association* 95.449 (2000), pp. 135–143.

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Generalized MIS: Schemes with Replacement

- Example with $N = 3$ proposals. For each $n \in \{1, 2, 3\}$:⁵
 - simulate $j_n \sim \text{Cat}([1, 2, 3]; [1/3, 1/3, 1/3])$
 - simulate $\mathbf{x}_n \sim q_{j_n}(\mathbf{x})$
 - weight $W_n = \frac{\pi(\mathbf{x}_n)}{\varphi_n(\mathbf{x}_n)}$ (several options for φ_n)
- 1. and 2. are equivalent to mixture sampling:

$$\mathbf{x}_n \stackrel{i.i.d.}{\sim} \psi(\mathbf{x}) = \frac{1}{3} \sum_{n=1}^N q_n(\mathbf{x})$$

Available proposals				
Sampling	j_n			
	$\mathbf{x}_n$	$\mathbf{x}_1 \sim q_3$	$\mathbf{x}_2 \sim q_3$	$\mathbf{x}_3 \sim q_1$
Weighting options $w_n = \frac{\pi(\mathbf{x})}{\varphi_n(\mathbf{x})}$	R1	$\frac{\pi(\mathbf{x}_1)}{q_3(\mathbf{x}_1)}$	$\frac{\pi(\mathbf{x}_2)}{q_3(\mathbf{x}_2)}$	$\frac{\pi(\mathbf{x}_3)}{q_1(\mathbf{x}_3)}$
	R2	$\frac{\pi(\mathbf{x}_1)}{\frac{1}{3}(q_3(\mathbf{x}_1) + q_3(\mathbf{x}_1) + q_1(\mathbf{x}_1))}$	$\frac{\pi(\mathbf{x}_2)}{\frac{1}{3}(q_3(\mathbf{x}_2) + q_3(\mathbf{x}_2) + q_1(\mathbf{x}_2))}$	$\frac{\pi(\mathbf{x}_3)}{\frac{1}{3}(q_3(\mathbf{x}_3) + q_3(\mathbf{x}_3) + q_1(\mathbf{x}_3))}$
	R3	$\frac{\pi(\mathbf{x}_1)}{\frac{1}{3}(q_1(\mathbf{x}_1) + q_2(\mathbf{x}_1) + q_3(\mathbf{x}_1))}$	$\frac{\pi(\mathbf{x}_2)}{\frac{1}{3}(q_1(\mathbf{x}_2) + q_2(\mathbf{x}_2) + q_3(\mathbf{x}_2))}$	$\frac{\pi(\mathbf{x}_3)}{\frac{1}{3}(q_1(\mathbf{x}_3) + q_2(\mathbf{x}_3) + q_3(\mathbf{x}_3))}$

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Generalized MIS: Schemes with No Replacement

- Example with $N = 3$ proposals. For each $n \in \{1, 2, 3\}$:⁶
 1. set $j_n = n$
 2. simulate $x_n \sim q_{j_n}(\mathbf{x})$
 3. weight $W_n = \frac{\pi(\mathbf{x}_n)}{\varphi_n(\mathbf{x}_n)}$ (several options for φ_n)
- 1. and 2. can be seen as mixture sampling $\mathbf{x}_n \sim \psi(\mathbf{x}) = \frac{1}{3} \sum_{n=1}^N q_n(\mathbf{x})$
 - but not i.i.d.!
 - only possible for mixtures with specific weights

Available proposals		①	②	③
Sampling	j_n	①	②	③
	$\mathbf{x}_n$	$\mathbf{x}_1 \sim q_1$	$\mathbf{x}_2 \sim q_2$	$\mathbf{x}_3 \sim q_3$
Weighting options	N1	$\frac{\pi(\mathbf{x}_1)}{q_1(\mathbf{x}_1)}$	$\frac{\pi(\mathbf{x}_2)}{q_2(\mathbf{x}_2)}$	$\frac{\pi(\mathbf{x}_3)}{q_3(\mathbf{x}_3)}$
	N3	$\frac{\pi(\mathbf{x}_1)}{\frac{1}{3}(q_1(\mathbf{x}_1)+q_2(\mathbf{x}_1)+q_3(\mathbf{x}_1))}$	$\frac{\pi(\mathbf{x}_2)}{\frac{1}{3}(q_1(\mathbf{x}_2)+q_2(\mathbf{x}_2)+q_3(\mathbf{x}_2))}$	$\frac{\pi(\mathbf{x}_3)}{\frac{1}{3}(q_1(\mathbf{x}_3)+q_2(\mathbf{x}_3)+q_3(\mathbf{x}_3))}$

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Theorem.⁷ For any **target distribution** $\pi(\mathbf{x})$, any integrable **function** h , and any **set of proposal densities** $\{q_n(\mathbf{x})\}_{n=1}^N$ such that the variance of the corresponding MIS estimators is finite,

$$\text{Var}(\hat{I}_{N1}) = \text{Var}(\hat{I}_{R1}) \geq \text{Var}(\hat{I}_{R3}) \geq \text{Var}(\hat{I}_{N3})$$

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Generalized MIS: Numerical example

- Simulate M total samples from two proposals:

$$\mathbf{x}_1^{(m)} \sim q_1, \quad m = 1, \dots, M/2$$

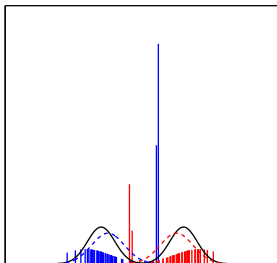
$$\mathbf{x}_2^{(m)} \sim q_2, \quad m = 1, \dots, M/2$$

- Weighting:

(a) **N1**

$$w_1^{(m)} = \frac{\pi(\mathbf{x}_1^{(m)})}{q_1(\mathbf{x}_1^{(m)})}$$

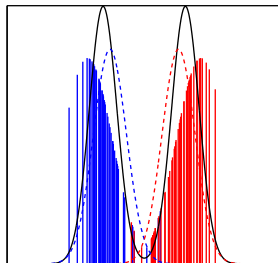
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$$w_1^{(m)} = \frac{\pi(\mathbf{x}_1)}{\frac{1}{2}(q_1(\mathbf{x}_1^{(m)}) + q_2(\mathbf{x}_1^{(m)}))}$$

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MIS Scheme	R1	N1	R2	N2	R3	N3
$\text{Var}(\hat{Z})$	1847	6874	10285	5474	0.01	0.01

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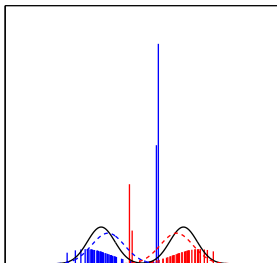
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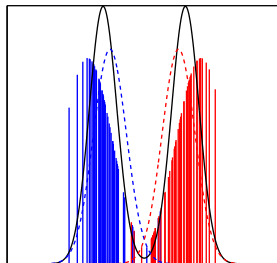
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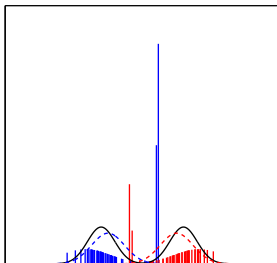
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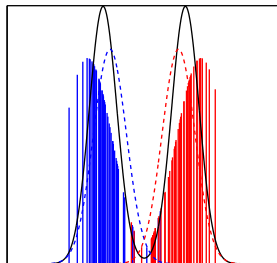
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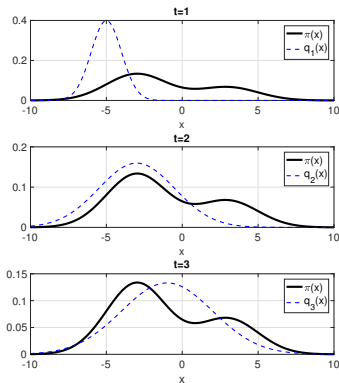
Adaptive Importance Sampling: Basics

- N proposals $\{q_{n,j}(\mathbf{x}|\boldsymbol{\theta}_{n,j})\}_{n=1}^N$ adapted over $j = 1, \dots, J$ iterations

$$\{q_{n,1}(\mathbf{x}|\boldsymbol{\theta}_{n,1})\}_{n=1}^N \rightarrow \{q_{n,2}(\mathbf{x}|\boldsymbol{\theta}_{n,2})\}_{n=1}^N \rightarrow \dots \rightarrow \{q_{n,J}(\mathbf{x}|\boldsymbol{\theta}_{n,J})\}_{n=1}^N$$

- Parametric AIS summarized as:⁸

$$\{\boldsymbol{\theta}_{n,1}\}_{n=1}^N \rightarrow \{\boldsymbol{\theta}_{n,2}\}_{n=1}^N \rightarrow \dots \rightarrow \{\boldsymbol{\theta}_{n,J}\}_{n=1}^N$$



⁸M. F. Bugallo et al. "Adaptive importance sampling: The past, the present, and the future". In: *IEEE Signal Processing Magazine* 34.4 (2017), pp. 60–79.

Adaptive Importance Sampling: Generic Algorithm

Initialization: Choose J , N , K , $\{q_{n,1}\}_{n=1}^N$, and initial parameters $\{\theta_{n,1}\}_{n=1}^N$

For $j = 1, \dots, J$:

1. **Sampling:** Simulate NK samples as

$$\mathbf{x}_{n,j}^{(k)} \sim q_{n,j}(\mathbf{x}|\theta_{n,j}), \quad k = 1, \dots, K, \quad n = 1, \dots, N.$$

2. **Weighting:** Weight the NK samples as [Elvira19]

$$W_{n,j}^{(k)} = \frac{\pi(\mathbf{x}_{n,j}^{(k)})}{\varphi_{n,j}(\mathbf{x}_{n,j}^{(k)})}, \quad k = 1, \dots, K, \quad n = 1, \dots, N.$$

3. **Adaptation of the parameters:** Update the proposal parameters

$$\{\theta_{n,j}\}_{n=1}^N \xrightarrow{\text{Adapt}} \{\theta_{n,j+1}\}_{n=1}^N$$

Outputs: NKJ weighted samples, $\{\mathbf{x}_{n,j}^{(k)}, W_{n,j}^{(k)}\}_{n=1, k=1, j=1}^{N, K, J}$

two questions: (1) **Weighting scheme?**⁹ (2) **Adaptive procedure of $\theta_{n,j}$?**¹⁰

⁹V. Elvira, L. Martino, D. Luengo, and M. F. Bugallo. "Generalized Multiple Importance Sampling". In: *Statistical Science* 34.1 (2019), pp. 129–155.

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Importance sampling for sequential inference

- Back to our problem: compute the joint $p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t})$ and/or filtering $p(\mathbf{x}_t|\mathbf{y}_{1:t})$:

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = p(\mathbf{x}_{1:t}, \mathbf{y}_{1:t})/Z$$

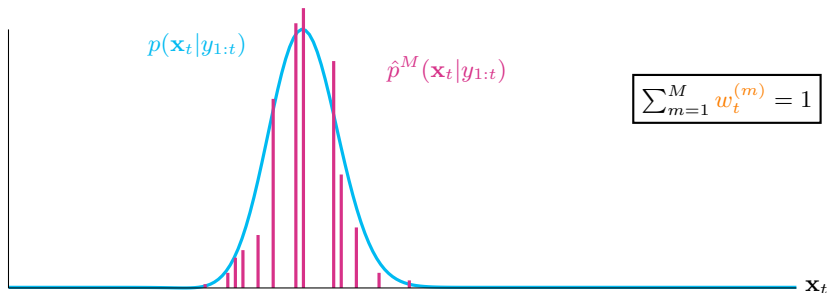
- Norm. constant: $Z = p(\mathbf{y}_{1:t}) = \int p(\mathbf{x}_{1:t}, \mathbf{y}_{1:t}) d\mathbf{x}_{1:t}$ cannot be computed
 - Marginalization: $p(\mathbf{x}_t|\mathbf{y}_{1:t}) = \int p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) d\mathbf{x}_{1:t-1}$ cannot be computed
- Importance sampling :
 - Sample M trajectories $\mathbf{x}_{1:t}^{(m)} \sim q(\mathbf{x}_{1:t})$, $m = 1, \dots, M$.
 - Weight each trajectory $W^{(m)} = \frac{p(\mathbf{x}_{1:t}^{(m)}|\mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})} \propto \frac{p(\mathbf{x}_{1:t}^{(m)}, \mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})}$, $m = 1, \dots, M$.
- Approximate the joint target with weighted trajectories

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

where $w^{(m)} = \frac{W^{(m)}}{\sum_{j=1}^M W^{(j)}}$ are the normalized weights ($\sum_{j=1}^M w^{(j)} = 1$).

Particle filtering/sequential Monte Carlo

- Particle filtering/sequential Monte Carlo
- The distributions are approximated by a random measure of M particles and associated normalized weights $\mathcal{X} = \{\mathbf{x}_t^{(m)}, w_t^{(m)}\}_{m=1}^M$
 - $p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx \hat{p}^M(\mathbf{x}_t | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t)$



Batch importance sampling

- Which proposal $q(\mathbf{x}_{1:t})$ for sampling/simulating the M trajectories $\mathbf{x}_{1:t}^{(n)}$?
 - Easiest choice: $q(\mathbf{x}_{1:t}) = p(\mathbf{x}_{1:t}) \equiv p(\mathbf{x}_1) \cdot p(\mathbf{x}_2|\mathbf{x}_1) \cdot \dots \cdot p(\mathbf{x}_t|\mathbf{x}_{t-1})$
 - * proposal factorizes (it is a choice)
- Batch procedure of IS with M samples:

1: **for** $m = 1$ to M **do**

2: 1. Sample the m -th trajectory (**sampling**):

$$\mathbf{x}_1^{(m)} \sim p(\mathbf{x}_1)$$

$$\mathbf{x}_2^{(m)} \sim p(\mathbf{x}_2|\mathbf{x}_1^{(m)})$$

...

$$3: \quad \mathbf{x}_{1:t}^{(m)} = [\mathbf{x}_1^{(m)}, \dots, \mathbf{x}_t^{(m)}] \quad \mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(m)})$$

4: 2. Weight for the m -th trajectory (**weighting**):

$$W_t^{(m)} = \frac{p(\mathbf{x}_{1:t}^{(m)}, \mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})} = \frac{p(\mathbf{y}_{1:t}|\mathbf{x}_{1:t}^{(m)})p(\mathbf{x}_{1:t}^{(m)})}{p(\mathbf{x}_{1:t}^{(m)})} = p(\mathbf{y}_{1:t}|\mathbf{x}_{1:t}^{(m)})$$

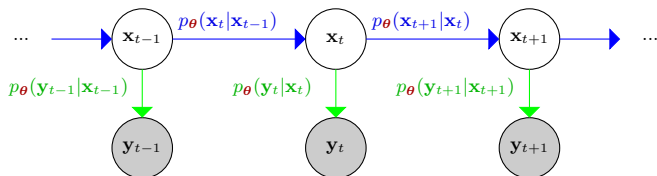
(likelihood evaluated at the m -th trajectory)

5: Normalize weights as $w^{(m)} = \frac{W_t^{(m)}}{\sum_{j=1}^M W_t^{(j)}}$

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

$$p(\mathbf{x}_t|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_t|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t) \quad (\text{Monte Carlo marginalization})$$

Sequential importance sampling (SIS)



- Due to model structure, joint likelihood factorizes as

$$w_t^{(m)} \propto p(\mathbf{y}_{1:t}|\mathbf{x}_{1:t}^{(m)}) = p(\mathbf{y}_1|\mathbf{x}_1^{(m)}) \cdot p(\mathbf{y}_2|\mathbf{x}_2^{(m)}) \cdot \dots \cdot p(\mathbf{y}_t|\mathbf{x}_t^{(m)})$$

- Towards sequential processing:
 - If we receive $\mathbf{y}_{t+1}$, can we approximate $p(\mathbf{x}_{1:t+1}|\mathbf{y}_{1:t+1})$ without re-processing $\mathbf{y}_{1:t}$?

$$\begin{aligned} p(\mathbf{x}_{1:t+1}|\mathbf{y}_{1:t+1}) &= \frac{p(\mathbf{y}_{t+1}, \mathbf{x}_{1:t+1}|\mathbf{y}_{1:t})}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} = \frac{p(\mathbf{y}_{t+1}|\mathbf{x}_{1:t+1}, \mathbf{y}_{1:t})}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} p(\mathbf{x}_{1:t+1}|\mathbf{y}_{1:t}) \quad (\text{Bayes}) \\ &= \frac{p(\mathbf{y}_{t+1}|\mathbf{x}_{1:t+1}, \mathbf{y}_{1:t})}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} p(\mathbf{x}_{t+1}|\mathbf{x}_{1:t}, \mathbf{y}_{1:t}) p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \quad (\text{factorization}) \\ &= \frac{p(\mathbf{y}_{t+1}|\mathbf{x}_{t+1}) p(\mathbf{x}_{t+1}|\mathbf{x}_t)}{p(\mathbf{y}_{t+1}|\mathbf{y}_{1:t})} p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \quad (\text{model structure}) \end{aligned}$$

Sequential importance sampling (SIS)

- Sequential importance sampling (SIS):
 - sample and weight sequentially at time t
 - process each observation $\mathbf{y}_t$ without reprocessing $\mathbf{y}_{1:t-1}$
- for** $t = 2$ to T **do**
 - for** $m = 1$ to M **do**
 1. Sample the m -th trajectory at time t :

$$\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})$$

- $\mathbf{x}_{1:t}^{(m)} = [\mathbf{x}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$

2. Weight for the m -th trajectory:

$$W_t^{(m)} = p(\mathbf{y}_{1:t} | \mathbf{x}_{1:t}^{(m)}) = W_{t-1}^{(m)} p(\mathbf{y}_t | \mathbf{x}_t^{(m)})$$

- end for**

- end for**

- Normalize weights: $w_T^{(m)} = \frac{W_T^{(m)}}{\sum_{j=1}^M W_T^{(j)}}$, $m = 1, \dots, M$.

- return** M trajectories with their weights: $\{\mathbf{x}_{1:T}^{(m)}, w_T^{(m)}\}_{m=1}^M$

Quality of the approximation and resampling step

- We can approximate any function f of the desired $p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t})$ with the self-normalized estimator:

$$\tilde{I}_N(f) = \sum_{m=1}^M f(\mathbf{x}_{1:t}^{(m)}) w_t^{(m)}$$

- Example: mean of the filtering distribution $p(\mathbf{x}_t|\mathbf{y}_{1:t})$ is the weighted average of the particles at time t .

$$\tilde{I}_N = \sum_{m=1}^M \mathbf{x}_t^{(m)} w_t^{(m)}$$

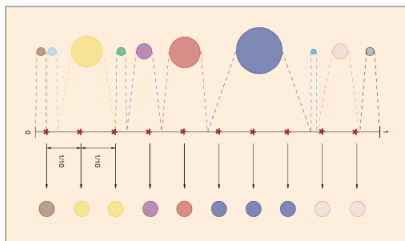
- The quality of the approximation depends on weights variability:
 - ★ if one particle gets a weight ≈ 1 (and the others almost zero), the approximation is **very bad**.

- **Resampling step:**

- at each time t we **kill bad trajectories** with very **low weight** and **replicate good trajectories** (before processing future observations)
- implicit improvement of the marginal proposal for $t + 1$

Resampling step

- Resampling step: third step (after 1. Sampling, and 2. Weighting)
 - easy but **necessary** to make the PF work.
- At time t , after computing the weights we **replicate good particles** and **kill bad particles**.
 - Urn example: draw i.i.d. M balls (particles) with replacement, with probability equal to the associated weights.
 - More precisely: $\mathcal{X}_t = \{\mathbf{x}_t^{(m)}, w_t^{(m)}\}_{m=1}^M$ forms an empirical distribution $p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$.
- * Sample M particles, $\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t})$, $m = 1, \dots, M$.



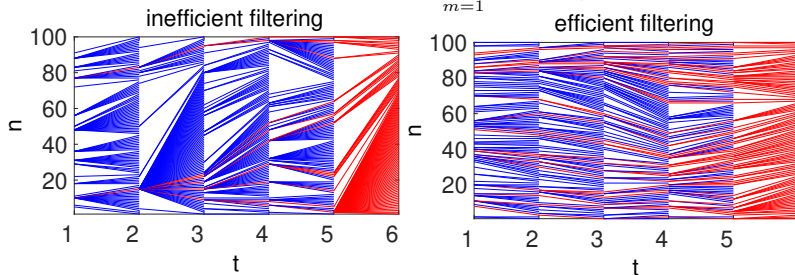
(Source in ¹¹)

¹¹P. M. Djuric et al. "Particle filtering". In: *IEEE signal processing magazine* 20.5 (2003), pp. 19–38.

Side effect of the resampling step

- Resampling:
 - reduces particle degeneracy (few samples get most of the probability mass)
 - side effect of introducing path degeneracy (few ancestor particles surviving)
- recall we approximate the joint target with weighted trajectories

$$p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) = \sum_{m=1}^M w^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$



- Resampling only when particle degeneracy is severe:
 - Effective sampling size (ESS):¹² $\widehat{\text{ESS}} = \frac{1}{\sum_{m=1}^M [w_t^{(m)}]^2}$
 - Adaptive resampling: resample only if $\widehat{\text{ESS}} < \gamma$, with $1 \leq \gamma \leq M$
 - * $\gamma = M$: resample always (equiv. BPF)
 - * $\gamma = 1$: never resample (equiv. batch IS)

¹²V. Elvira, L. Martino, and C. P. Robert. "Rethinking the effective sample size". In: *International Statistical Review* 90.3 (2022), pp. 525–550.

The bootstrap PF (BPF)

- Bootstrap PF $\equiv$ Sequential importance sampling resampling (SISR)¹³

(i) Initialization. At time $t = 0$, $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$, $m = 1, \dots, M$.

(ii) Recursive step. At time t ,

1. **Prediction** (particles propagation): $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$

2. **Update** (weights calculation): compute the normalized weights as $w_t^{(m)} \propto p(\mathbf{y}_t | \mathbf{x}_t^{(m)})$ associated to trajectory $\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$

3. **Multinomial resampling**

a) simulate $i^{(m)} \sim \text{Cat}([1, \dots, M]; [w_t^{(1)}, \dots, w_t^{(M)}])$, $m = 1, \dots, M$

b) set $\tilde{\mathbf{x}}_{1:t}^{(m)} = \mathbf{x}_{1:t}^{(i^{(m)})}$, $m = 1, \dots, M$

equivalent to simulate M i.i.d. samples from the approx. filtering dist.

$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

$$p(\mathbf{x}_{1:t} | \mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_{1:t} | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_{1:t}^{(m)}}(\mathbf{x}_{1:t})$$

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) = \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t) \approx \frac{1}{M} \sum_{m=1}^M \delta_{\tilde{\mathbf{x}}_t^{(m)}}(\mathbf{x}_t)$$

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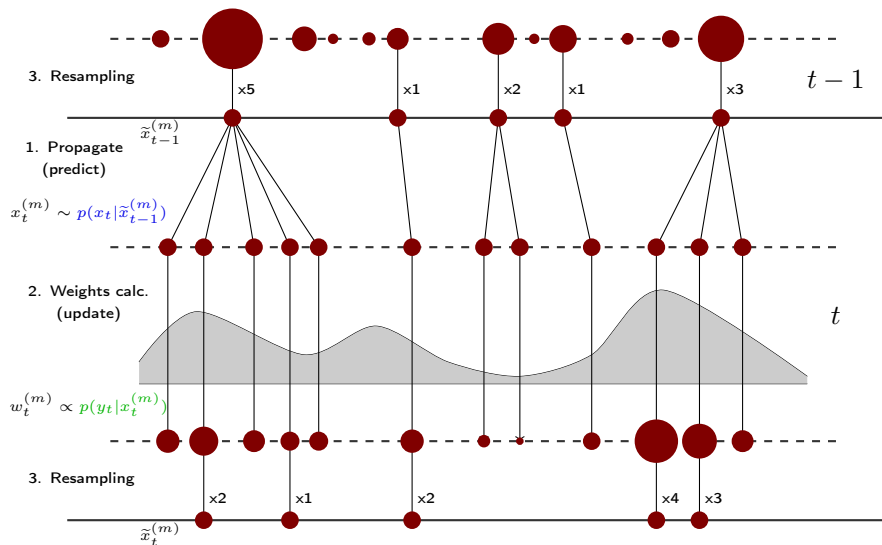
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Bootstrap particle filter



Outline

State-space models and Bayesian filtering

Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

Particle filtering

- From IS to sequential IS

Mini-project

- In search of better particle filters

- Mini-project

Particle filtering from the MIS and AIS perspectives

Advanced particle filtering

- PFs in high-dimension spaces

- The choice of N in PFs

Mini-project: BF for the Lorenz 63 model

- Take the same state-space model as for the Kalman filtering problem (stochastic Lorenz 63, with Euler discretisation and nonlinear observations) and code a standard bootstrap filter with N particles.
- Compare the performance of the bootstrap filter and the non-linear extensions of KF (e.g., CKF): try different values of state noise variance and observational noise variance, gap between observations, and M (number of particles).

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Beyond BPF

- The m -th proposal in BPF is the transition kernel $q(\mathbf{x}_t) = p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$
 - note that is conditioned on $\tilde{\mathbf{x}}_{t-1}^{(m)}$ which comes from a resampling step.
 - proposal of all samples can be interpreted as approximate predictive, since resampling $(t-1)$ + propagation at t = mixture sampling:

$$\mathbf{x}_t^{(m)} \stackrel{i.i.d.}{\sim} p^M(\mathbf{x}_t | \mathbf{y}_{1:t-1}) = \sum_{m=1}^M w_t^{(m)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})$$

- From IS theory, high variance of the importance weights $\Rightarrow$ low efficiency/accuracy of the filter
- **Intuition** (imprecise): inefficiency in BPF will happen when predictive $p(\mathbf{x}_t | \mathbf{y}_{1:t-1})$, thus mixture proposal, differs from **filtering distribution**:

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) = \frac{p(\mathbf{y}_t | \mathbf{x}_t) p(\mathbf{x}_t | \mathbf{y}_{1:t-1})}{p(\mathbf{y}_t | \mathbf{y}_{1:t-1})}$$

- equivalent to say that $\mathbf{y}_t$ is **informative**, which happens when:
 - * $p(\mathbf{y}_t | \mathbf{x}_t)$ is "peaky" compared to $p(\mathbf{x}_t | \mathbf{y}_{1:t-1})$ (e.g., very low-variance observation noise)
 - * $p(\mathbf{y}_t | \mathbf{x}_t)$ is placed in a "different" area of the space compared to $p(\mathbf{x}_t | \mathbf{y}_{1:t-1})$ (e.g., outlier observation)
- in those scenarios, great variability of the weights.

An adapted PF with generic proposal

- A good proposal must include knowledge about $\mathbf{y}_t$
 - avoid particles being sampled into regions of the state space which are unlikely in light of that observation
- Generic sampling:

$$\mathbf{x}_{1:t}^{(m)} \sim q(\mathbf{x}_{1:t}) = \prod_{k=1}^t q(\mathbf{x}_k; \mathbf{x}_{1:k-1}, \mathbf{y}_{1:k})$$

with trajectory weight:

$$W_t^{(m)} = \frac{p(\mathbf{x}_{1:t}^{(m)}, \mathbf{y}_{1:t})}{q(\mathbf{x}_{1:t}^{(m)})} = \frac{p(\mathbf{y}_{1:t} | \mathbf{x}_{1:t}^{(m)}) p(\mathbf{x}_{1:t}^{(m)})}{q(\mathbf{x}_{1:t}^{(m)})} = \frac{\prod_{k=1}^t p(\mathbf{y}_k | \mathbf{x}_k^{(m)}) p(\mathbf{x}_k^{(m)} | \mathbf{x}_{k-1}^{(m)})}{\prod_{k=1}^t q(\mathbf{x}_k^{(m)}; \mathbf{x}_{1:k-1}^{(m)}, \mathbf{y}_{1:k})}$$

- possible proposal choices:
 - BPF (online-oriented) uninformative:

$$\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \mathbf{x}_{1:t-1}^{(m)}, \mathbf{y}_{1:t}) = q(\mathbf{x}_t; \mathbf{x}_{t-1}^{(m)}) = p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})$$

- still convenient (online-oriented) but more informative:

$$\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \mathbf{x}_{1:t-1}^{(m)}, \mathbf{y}_{1:t}) = q(\mathbf{x}_t; \mathbf{x}_{t-1}^{(m)}, \mathbf{y}_t)$$

An adapted PF with generic proposal

- (i) Initialization. At time $t = 0$, $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$, $m = 1, \dots, M$.
- (ii) Recursive step. At time t ,
1. **Sampling/propagation:** $\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)$ associated to trajectory $\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$
 2. **Weights calculation:** compute the normalized weights as
$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$
 3. **Multinomial resampling**
 - a) simulate $i^{(m)} \sim \text{Cat}([1, \dots, M]; [w_t^{(1)}, \dots, w_t^{(M)}])$, $m = 1, \dots, M$
 - b) set $\tilde{\mathbf{x}}_t^{(m)} = \mathbf{x}_t^{(i^{(m)})}$, $m = 1, \dots, M$equivalent to simulate M i.i.d. samples from the approx. filtering dist.

$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

An adapted PF with generic proposal

(i) Initialization. At time $t = 0$, $\tilde{\mathbf{x}}_0^{(m)} \sim p(\mathbf{x}_0)$, $m = 1, \dots, M$.

(ii) Recursive step. At time t ,

1. **Sampling/propagation:** $\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)$ associated to trajectory

$$\mathbf{x}_{1:t}^{(m)} = [\tilde{\mathbf{x}}_{1:t-1}^{(m)}, \mathbf{x}_t^{(m)}]$$

2. **Weights calculation:** compute the normalized weights as

$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$

3. **Multinomial resampling**

a) simulate $i^{(m)} \sim \text{Cat}([1, \dots, M]; [w_t^{(1)}, \dots, w_t^{(M)}])$, $m = 1, \dots, M$

b) set $\tilde{\mathbf{x}}_t^{(m)} = \mathbf{x}_t^{(i^{(m)})}$, $m = 1, \dots, M$

equivalent to simulate M i.i.d. samples from the approx. filtering dist.

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An adapted PF with generic proposal

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$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$
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$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

An adapted PF with generic proposal

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$$\tilde{\mathbf{x}}_t^{(m)} \sim p^M(\mathbf{x}_t | \mathbf{y}_{1:t}) \equiv \sum_{j=1}^M w_t^{(j)} \delta_{\mathbf{x}_t^{(j)}}(\mathbf{x})$$

In search of “optimal” filters

- **Proposal choice** to minimize variance of

$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \tilde{\mathbf{x}}_{t-1}^{(m)})}{q(\mathbf{x}_t^{(m)}; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t)}$$

- “optimal” kernel: $q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}^{(m)}, \mathbf{y}_t) = p(\mathbf{x}_t | \mathbf{y}_t, \tilde{\mathbf{x}}_{t-1}^{(m)})$
 - * proportional to the numerator $p(\mathbf{y}_t, \mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$
 - * intractable kernel
 - * reduces the variance of each weight (constant) but still weights are different across them:

$$W_t^{(m)} = p(\mathbf{y}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$$

- Even if available, the kernel does not solve all the problems: the resampling remains blind to the new observation $\mathbf{y}_t$
 - **Goal:** we would like to modify the resampling weights to replicate trajectories at $t - 1$ that will perform better in t

In search of “optimal” filters: trajectory perspective

- Recap: BPF and adapted PF proceed in the following order:
 - Resampling ($t - 1$): resample trajectories $\tilde{\mathbf{x}}_{1:t-1}^{(n)}$ ($\mathbf{y}_t$ is **not used** in adapted PF nor in BPF)
 - ★ recall: equivalent to simulating at $t - 1$ as
$$\tilde{\mathbf{x}}_{1:t-1}^{(m)} \sim p^M(\mathbf{x}_{1:t-1} | \mathbf{y}_{1:t-1}) \equiv \sum_{j=1}^M w_{t-1}^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1})$$
 - Sampling (t): propagate from $\tilde{\mathbf{x}}_{t-1}^{(m)}$ to $\mathbf{x}_t^{(m)}$ ($\mathbf{y}_t$ is **used in adapted PF** but **not used in BPF**)
 - Weighting (t): Bayesian update ($\mathbf{y}_t$ is used)
- Is it possible to use $\mathbf{y}_t$ also at resampling?

Filter	Sampling	Resampling
Bootstrap PF	✗	✗
Adapted PF	✓	✗
Optimal PF	✓	✓

In search of better filters: trajectory perspective

- More precisely, resample trajectories from $p^M(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t})$ instead of $p^M(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t-1})$?

$$\begin{aligned}
 p(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t}) &= \int p(\mathbf{x}_{1:t}|\mathbf{y}_{1:t}) d\mathbf{x}_t \\
 &= \int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})p(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t-1})d\mathbf{x}_t \\
 &= p(\mathbf{x}_{1:t-1}|\mathbf{y}_{1:t-1}) \int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})d\mathbf{x}_t \\
 &\approx \left[\sum_{j=1}^M w_{t-1}^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1}) \right] \int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1})d\mathbf{x}_t \\
 &= \sum_{j=1}^M w_{t-1}^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1}) \underbrace{\int p(\mathbf{y}_t|\mathbf{x}_t)p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(j)})d\mathbf{x}_t}_{v_t^{(j)}=p(\mathbf{y}_t|\mathbf{x}_{t-1}^{(j)})} \\
 &= \sum_{j=1}^M \underbrace{w_{t-1}^{(j)} v_t^{(j)}}_{\lambda_t^{(j)}} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1})
 \end{aligned}$$

- $v_t^{(j)}$ is intractable $\Rightarrow$ cheap approximation: $v_t^{(j)} \approx p(\mathbf{y}_t|\bar{\mathbf{x}}_t^{(j)})$
 - justification: $p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(j)}) \approx \delta_{\bar{\mathbf{x}}_t^{(j)}}(\mathbf{x}_t)$, where $\bar{\mathbf{x}}_t^{(j)} = \mathbb{E}_{p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(j)})}[\mathbf{x}_t]$

In search of better filters: trajectory perspective

New approach:

1. Resampling ($t - 1$): resample trajectories $\tilde{\mathbf{x}}_{1:t-1}^{(m)}$ ($\mathbf{y}_t$ is now **used**)
2. Sampling (t): simulate from $q(\mathbf{x}_t; \tilde{\mathbf{x}}_{t-1}, \mathbf{y}_t)$ ($\mathbf{y}_t$ potentially **used**)
 - trajectory proposal is now

$$q(\mathbf{x}_t; \mathbf{x}_{1:t-1}) = q(\mathbf{x}_t; \mathbf{x}_{t-1}, \mathbf{y}_t) \sum_{j=1}^M \lambda_t^{(j)} \delta_{\mathbf{x}_{1:t-1}^{(j)}}(\mathbf{x}_{1:t-1})$$

and sampling as

a) simulate $i^{(m)} \sim \text{Cat}([1, \dots, M]; [\lambda_t^{(1)}, \dots, \lambda_t^{(M)}])$, $m = 1, \dots, M$

b) simulate $\mathbf{x}_t^{(m)} \sim q(\mathbf{x}_t; \mathbf{x}_{t-1}^{(i^{(m)})}, \mathbf{y}_t)$, $m = 1, \dots, M$

3. Weighting (t) in the joint space, using $\mathbf{x}_{1:t-1}^{(j)}$ fulfills

$$\begin{aligned} q(\mathbf{x}_{1:t-1}^{(j)}; \mathbf{y}_{1:t}) &= \frac{\lambda_t^{(j)}}{w_t^{(j)}} p^M(\mathbf{x}_{1:t-1}^{(j)} | \mathbf{y}_{1:t-1}), \\ w_t(\mathbf{x}_{1:t}^{(m)}) &= \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^m | \mathbf{x}_{t-1}^{(i^{(m)})}) p(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t-1})}{q(\mathbf{x}_t^{(m)}; \mathbf{x}_{1:t-1}^{(i^{(m)})}, \mathbf{y}_t) q(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t})} \\ &= \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^m | \mathbf{x}_{t-1}^{(i^{(m)})}) p(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t-1})}{q(\mathbf{x}_t^{(m)}; \mathbf{x}_{1:t-1}^{(i^{(m)})}, \mathbf{y}_t) \frac{\lambda_t^{(i^{(m)})}}{w_t^{(i^{(m)})}} p^M(\mathbf{x}_{1:t-1}^{(i^{(m)})} | \mathbf{y}_{1:t-1})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^m | \mathbf{x}_{t-1}^{(i^{(m)})})}{q(\mathbf{x}_t^{(m)}; \mathbf{x}_{1:t-1}^{(i^{(m)})}, \mathbf{y}_t) p(\mathbf{y}_t | \mu_t^{(i^{(m)})})} \end{aligned}$$

Auxiliary PF (APF)

- Proposed in ¹⁴ as an alternative to BPF
 - APF improves sometimes the performance of BPF, but **not always**.
 - it attempts to sample in better areas in light of the new observation y_t
- (i) Initialization. At time $t = 0$, $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$, and $w_0^{(m)} = 1/M$, $m = 1, \dots, M$.
- (ii) Recursive step. At time $t > 0$,
 - Modify weights before resampling.** Compute

$$\bar{\mathbf{x}}_t^{(m)} = \mathbb{E}_{p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(m)})}[\mathbf{x}_t], \quad m = 1, \dots, M.$$

and the normalized weights ($\sum_{m=1}^M \lambda_t^{(m)} = 1$)

$$\lambda_t^{(m)} \propto p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(m)}) w_{t-1}^{(m)}, \quad m = 1, \dots, M,$$

- Delayed resampling.** Select the indexes $i^{(m)} = j$, with probability proportional to $\lambda_t^{(j)}$, $m = 1, \dots, M$
- Prediction.** $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(i^{(m)})})$, $m = 1, \dots, M$.
- Update.** Compute the normalized weights as

$$w_t^{(m)} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(i^{(m)})})}, \quad m = 1, \dots, M.$$

¹⁴M. K. Pitt and N. Shephard. "Filtering via simulation: Auxiliary particle filters". In: *Journal of the American statistical association* 94.446 (1999), pp. 590–599. > < ≡ ≡ ≡

Outline

State-space models and Bayesian filtering

Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

- Mini-project**

Particle filtering from the MIS and AIS perspectives

Advanced particle filtering

- PFs in high-dimension spaces

- The choice of N in PFs

Mini-project

- Assume, again, the stochastic Lorenz 63 model, time-discretised using an Euler-Maruyama scheme:

$$X_{1,n} = X_{1,n-1} - hs(X_{1,n-1} - X_{2,n-1}) + \sigma\sqrt{h}Z_{1,n},$$

$$X_{2,n} = X_{2,n-1} + h(rX_{1,n-1} - X_{2,n-1} - X_{1,n-1}X_{3,n-1}) + \sigma\sqrt{h}Z_{2,n},$$

$$X_{3,n} = X_{3,n-1} + h(X_{1,n-1}X_{2,n-1} - bX_{3,n-1}) + \sigma\sqrt{h}Z_{3,n},$$

where $Z_{i,n} \sim \mathcal{N}(0, 1)$, the state is $X_n = [X_{1,n}, X_{2,n}, X_{3,n}]^\top$ and the parameters are $(s, r, b) = (10, 28, \frac{8}{3})$. You may 'play around' with the value of σ as in the previous mini-projects (start with $\sigma = \frac{1}{2}$).

- Assume linear observations, namely $Y_n = X_{1,n} + \sigma_u U_n$, where $U_n \sim \mathcal{N}(0, 1)$, every B discrete time steps (e.g., $B = 40$). Again, you may try different values of the parameter σ_u (you may start with $\sigma_u = 2$).

Mini-project

- For the state space model in the previous slide, code a standard SIR algorithm and a SIR algorithm with a Gaussian proposal.
Try different configurations of the model noise parameters (σ_u, σ, M) and plot the ground-truth signals and their estimates with both filters. Reference values:
 $\sigma_u = 2, \sigma = 1, B = 40$ and $M = 50$ particles in the SIR algorithm.
The simulation should also return estimation errors (e.g., average square errors over time) for the two filters.

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Revisiting standard filters from the MIS perspective

- Both BPF and APF (and other filters) use several proposals at each t
- e.g., BPF proposal is $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$
 - potentially M different values $\tilde{\mathbf{x}}_{t-1}^{(m)}$, i.e., M proposals.
 - at least two views:
 1. each sample $\mathbf{x}_t^{(m)}$ is simulated from $p(\mathbf{x}_t | \tilde{\mathbf{x}}_{t-1}^{(m)})$
 2. all samples are i.i.d. samples from the mixture

$$p^M(\mathbf{x}_t | \mathbf{y}_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)})$$

- Is it possible to re-interpret BPF and other filters from a MIS perspective?

A generic particle filtering from the MIS perspective

- (i) Initialization. At time $t = 0$, $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$, $w_0^{(m)} = 1/M$, $m = 1, \dots, M$.
- (ii) Recursive step. At time $t > 0$,

1 **Proposal adaptation/selection.** Select the MIS proposal of the form

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}),$$

2 **Sampling.** Draw samples according to

$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M.$$

3 **Weighting.** Compute the normalized IS weights by

$$\begin{aligned} w_t^{(m)} &\propto \frac{p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t})}{\psi_t(\mathbf{x}_t^{(m)})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t-1})}{\psi_t(\mathbf{x}_t^{(m)})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\psi_t(\mathbf{x}_t^{(m)})} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \end{aligned} \quad (2)$$

• Two questions:¹⁵

1. Selection/adaptation of $\{\lambda_t^{(j)}\}_{j=1}^M$ to build $\psi_t(\mathbf{x}_t)$?

* Recall: IS is efficient when $\psi_t(\mathbf{x}_t)$ is close to $p(\mathbf{x}_t | \mathbf{y}_{1:t}) \Rightarrow$ AIS

2. Approximate $w_t^{(m)}$ in (2) to derive BPF, APF. and other/new filters?

¹⁵V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

BPF from the MIS perspective

(i) Initialization. At time $t = 0$, $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$, and $w_0^{(m)} = 1/M$, $m = 1, \dots, M$.

(ii) Recursive step. At time $t > 0$,

1 **Proposal adaptation/selection.** Select the MIS proposal of the form

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}), \quad (\lambda_t^{(j)} = w_{t-1}^{(j)})$$

2 **Sampling.** Draw samples according to

$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M. \quad (\text{equiv. resampling+propagation})$$

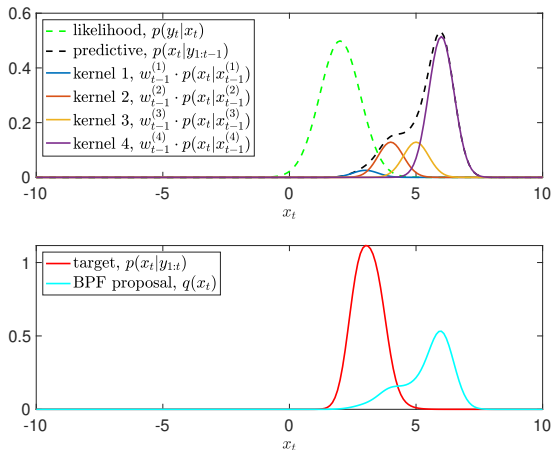
3 **Weighting.** Compute the normalized IS weights by

$$\begin{aligned} w_t^{(m)} &\propto \frac{p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t})}{\psi_t(\mathbf{x}_t^{(m)})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t-1})}{\psi_t(\mathbf{x}_t^{(m)})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\psi_t(\mathbf{x}_t^{(m)})} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \\ &= p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \end{aligned}$$

• Remark: the BPF proposal matches just the prior of the numerator.¹⁶

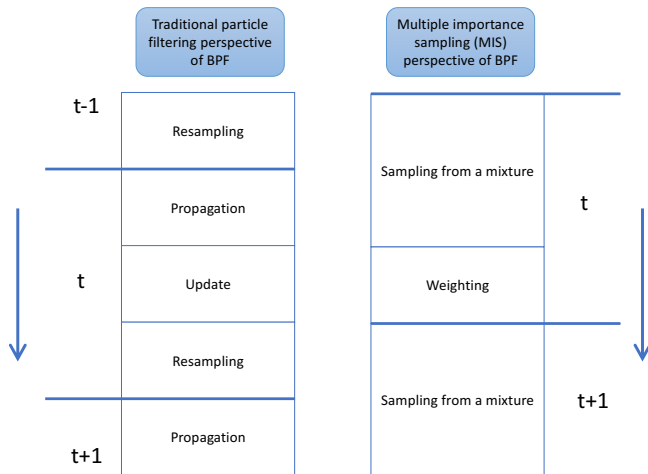
¹⁶V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

Toy example: BPF with $M = 4$ particles



- predictive, $p(x_t|y_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(x_t|x_{t-1}^{(j)})$ with $w_{t-1} = [0.03, 0.16, 0.16, 0.65]$
- BPF proposal, $\psi_t^{\text{BPF}}(x_t) = \sum_{j=1}^M \lambda_t^{(j)} p(x_t|x_{t-1}^{(j)})$, with $\lambda_t^{\text{BPF}} = w_{t-1}^{(m)} = [0.03, 0.16, 0.16, 0.65]$

BPF from the MIS perspective



APF from the MIS perspective

(i) Initialization. At time $t = 0$, $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$, and $w_0^{(m)} = 1/M$, $m = 1, \dots, M$.

(ii) Recursive step. At time $t > 0$,

1 **Proposal adaptation/selection.** The weight of each mixture kernel is amplified by the likelihood eval. at its center $\bar{\mathbf{x}}_t^{(m)} = \mathbb{E}_{p(\mathbf{x}_t|\mathbf{x}_{t-1}^{(m)})}[\mathbf{x}_t]$, i.e.,

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}), \quad \text{with } \lambda_t^{(j)} \propto p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(j)}) w_{t-1}^{(j)}, \quad j = 1, \dots, M$$

2 **Sampling.** Draw M i.i.d. samples from the mixture $\psi_t(\mathbf{x}_t)$, i.e.,

a) Select the indexes $i^{(m)} = j$, with probability $\propto \lambda_t^{(j)}$, $m = 1, \dots, M$

b) simulate $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(i^{(m)})})$, $m = 1, \dots, M$.

3 **Weighting.** Compute the normalized IS weights by

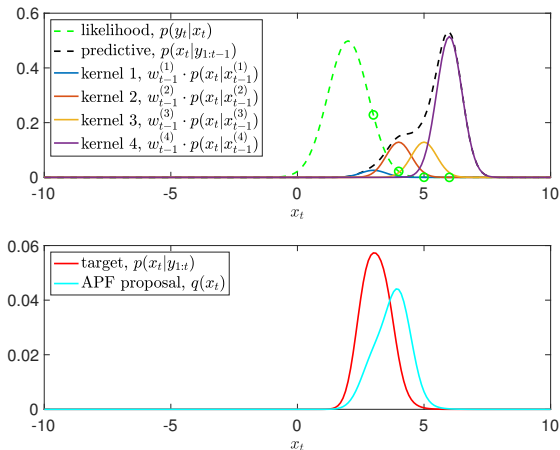
$$\begin{aligned} w_t^{(m)} &\propto \frac{p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t})}{\psi_t(\mathbf{x}_t^{(m)})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) p(\mathbf{x}_t^{(m)} | \mathbf{y}_{1:t-1})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})} \\ &\approx \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) w_{t-1}^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})}{\lambda_t^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})} \propto \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) w_{t-1}^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})}{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(i^{(m)})}) w_{t-1}^{(i^{(m)})} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(i^{(m)})})} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(i^{(m)})})} \end{aligned}$$

• Remark:¹⁷

- implicit assumption: kernels are far apart
- the APF re-weights the kernels of the prior amplifying them with the likelihood (each of them, independently from the rest).

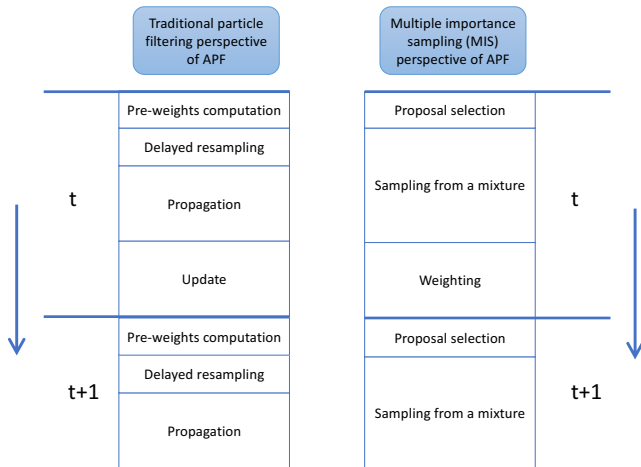
¹⁷V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

Toy example: APF with $M = 4$ particles



- predictive, $p(x_t|y_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(x_t|x_{t-1}^{(j)})$ with $w_{t-1} = [0.03, 0.16, 0.16, 0.65]$
- APF proposal, $\psi_t^{\text{APF}}(x_t) = \sum_{j=1}^M \lambda_t^{(j)} p(x_t|x_{t-1}^{(j)})$, with $\lambda_t^{\text{APF}} = p(y_t|\bar{x}_t^{(m)}) w_{t-1}^{(m)} = [0.6713, 0.3221, 0.0065, 0.0001]$

Auxiliary PF (APF) from the MIS perspective



Improved APF (IAPF)

- IAPF:¹⁸ Based on this MIS interpretation, we improve the APF
 - MIS perspective: the proposal is a **mixture of the same predictive kernels** as in BPF and APF

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)})$$

with

$$\lambda_t^{(j)} \propto \frac{p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(j)}) \sum_{k=1}^M w_{t-1}^{(k)} p(\bar{\mathbf{x}}_t^{(j)} | \mathbf{x}_{t-1}^{(k)})}{\sum_{k=1}^M p(\bar{\mathbf{x}}_t^{(j)} | \mathbf{x}_{t-1}^{(k)})}, \quad j = 1, \dots, M.$$

- **Interpretation:** the “amplification” $\lambda_t^{(j)}$ of j -th kernel, takes into account where all other kernels are placed (unlike APF)
 - * if kernels have small overlap, $\lambda_t^{(j)} \approx p(\mathbf{y}_t | \bar{\mathbf{x}}_t^{(j)}) w_{t-1}^{(j)}$ (IAPF reduces to APF)
- Connection to marginal PFs ¹⁹

¹⁸V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djurić. “In search for improved auxiliary particle filters”. In: *2018 26th European Signal Processing Conference (EUSIPCO)*. IEEE, 2018, pp. 1637–1641.

¹⁹M. Klaas, N. De Freitas, and A. Doucet. “Toward practical N2 Monte Carlo: The marginal particle filter”. In: *arXiv preprint arXiv:1207.1396* (2012).

Optimized APF (OAPF)

- OAPF:²⁰²¹ $K < M$ kernels form the mixture proposal
 - Optimized λ via non-negative least squares (NNLS) by taking the squared distance between target and mixture proposal at the E evaluation points

$$\boldsymbol{\lambda}^* = \arg \min_{\boldsymbol{\lambda}} \|\mathbf{Q}\boldsymbol{\lambda} - \tilde{\boldsymbol{\pi}}\|_2^2 \quad \text{subject to : } \boldsymbol{\lambda} \in \mathbb{R}_{\geq 0}^K.$$

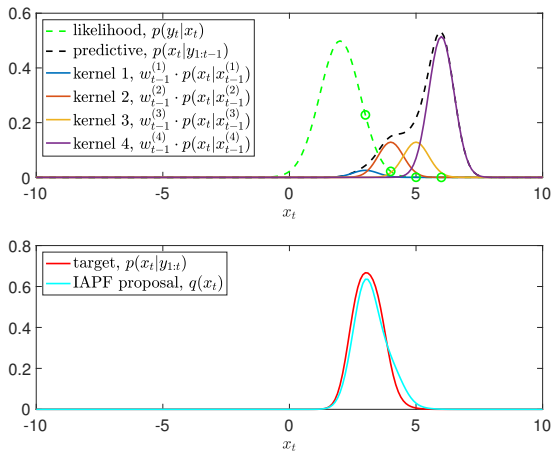
- In IAPF and OAPF, IS weights are *exact*:

$$w_t^{(m)} = \frac{p(\mathbf{y}_t | \mathbf{x}_t^{(m)}) \sum_{j=1}^K w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^K \lambda_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} | \mathbf{x}_{t-1}^{(j)})}$$

²⁰N. Branchini and V. Elvira. “Optimized auxiliary particle filters”. In: *Uncertainty in Artificial Intelligence*. PMLR. 2021, pp. 1289–1299.

²¹N. Branchini and V. Elvira. “An adaptive mixture view of particle filters”. In: *Foundations of Data Science* (2024), pp. 0–0.

Toy example: IAPF with $M = 4$ particles



- predictive, $p(x_t|y_{1:t-1}) = \sum_{j=1}^M w_{t-1}^{(j)} p(x_t|x_{t-1}^{(j)})$ with $w_{t-1} = [0.03, 0.16, 0.16, 0.65]$
- IAPF proposal, $\psi_t^{\text{IAPF}}(x_t) = \sum_{j=1}^M \lambda_t^{(j)} p(x_t|x_{t-1}^{(j)})$, with $\lambda_t^{\text{IAPF}} = [0.7657, 0.2276, 0.0066, 0.0001]$

Summary: PF framework from MIS perspective

- (i) Initialization. At time $t = 0$, $\mathbf{x}_0^{(m)} \sim p(\mathbf{x}_0)$, and $w_0^{(m)} = 1/M$, $m = 1, \dots, M$.
- (ii) Recursive step. At time $t > 0$,
- Proposal adaptation/selection.**²² Select the MIS proposal of the form

$$\psi_t(\mathbf{x}_t) = \sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t | \mathbf{x}_{t-1}^{(j)}), \quad \text{with} \quad \lambda_t^{(j)} = ?$$

- Sampling.** Draw samples according to

$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M.$$

- Weighting.** Compute the normalized IS weights by

$$w_t^{(m)} = ?$$

	BPF	APF	IAPF and OAPF
$\lambda_t^{(m)}$	$w_{t-1}^{(m)}$	$\propto p(\mathbf{y}_t \bar{\mathbf{x}}_t^{(m)}) w_{t-1}^{(m)}$	$\propto \frac{p(\mathbf{y}_t \bar{\mathbf{x}}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\bar{\mathbf{x}}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M p(\bar{\mathbf{x}}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}$
$w_t^{(m)}$	$\propto p(\mathbf{y}_t \mathbf{x}_t^{(m)})$	$\propto \frac{p(\mathbf{y}_t \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t \bar{\mathbf{x}}_t^{(m)})}$	$\propto \frac{p(\mathbf{y}_t \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}$

- In all PFs:

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t)$$

²²V. Elvira, L. Martino, M. F. Bugallo, and P. M. Djuric. "Elucidating the auxiliary particle filter via multiple importance sampling [lecture notes]". In: *IEEE Signal Processing Magazine* 36.6 (2019), pp. 145–152.

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$$\mathbf{x}_t^{(m)} \sim \psi_t(\mathbf{x}_t), \quad m = 1, \dots, M.$$

- Weighting.** Compute the normalized IS weights by

$$w_t^{(m)} = ?$$

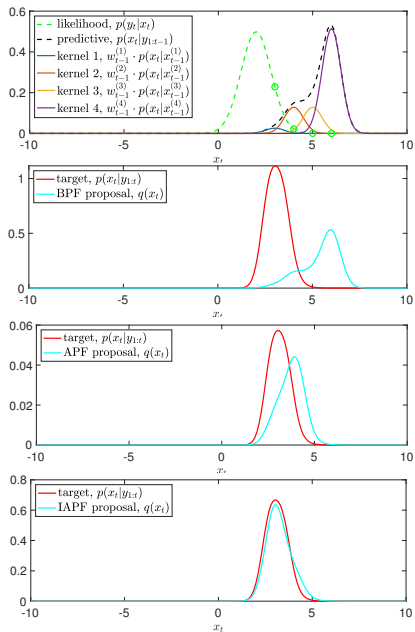
	BPF	APF	IAPF and OAPF
$\lambda_t^{(m)}$	$w_{t-1}^{(m)}$	$\propto p(\mathbf{y}_t \bar{\mathbf{x}}_t^{(m)}) w_{t-1}^{(m)}$	$\propto \frac{p(\mathbf{y}_t \bar{\mathbf{x}}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\bar{\mathbf{x}}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M p(\bar{\mathbf{x}}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}$
$w_t^{(m)}$	$\propto p(\mathbf{y}_t \mathbf{x}_t^{(m)})$	$\propto \frac{p(\mathbf{y}_t \mathbf{x}_t^{(m)})}{p(\mathbf{y}_t \bar{\mathbf{x}}_t^{(i_m)})}$	$\propto \frac{p(\mathbf{y}_t \mathbf{x}_t^{(m)}) \sum_{j=1}^M w_{t-1}^{(j)} p(\mathbf{x}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}{\sum_{j=1}^M \lambda_t^{(j)} p(\mathbf{x}_t^{(m)} \mathbf{x}_{t-1}^{(j)})}$

- In all PFs:

$$p(\mathbf{x}_t | \mathbf{y}_{1:t}) \approx \sum_{m=1}^M w_t^{(m)} \delta_{\mathbf{x}_t^{(m)}}(\mathbf{x}_t)$$

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Toy example: summary



Numerical result 1: channel estimation in wireless system

- We suppose a linear-Gaussian system described by

$$\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \mathbf{r}_t,$$

$$y_t = \mathbf{h}_t^\top \mathbf{x}_t + \mathbf{r}_t,$$

- $\mathbf{h}_t = [h_t, h_{t-1}, \dots, h_{t-d_x+1}]^\top$, last d_x transmitted pilots, $d_t \in \{-1, +1\}$,
- $\mathbf{A} = 0.7\mathbf{I}$
- $\mathbf{q}_t \sim \mathcal{N}(0, \mathbf{Q})$, $\mathbf{Q} = 5\mathbf{I}$
- $\mathbf{r}_t \sim \mathcal{N}(0, \mathbf{R})$, $\mathbf{R} = 0.5$
- we set $T = 200$ time steps and $M = 100$ particles

d_x (dimension)	1	2	3	5	10
MSE (BPF)	0.0272	0.3762	0.9657	1.4705	2.9592
MSE (APF)	0.0709	0.8041	1.6041	2.2132	3.7187
MSE (IAPF)	0.0062	0.1764	0.5176	0.8041	2.6931

Outline

State-space models and Bayesian filtering

Importance sampling: basics and advanced methods

- Monte Carlo basics

- Importance sampling (IS) basics

- Multiple importance sampling (MIS)

- Adaptive importance sampling (AIS)

Particle filtering

- From IS to sequential IS

- Mini-project

- In search of better particle filters

- Mini-project

Particle filtering from the MIS and AIS perspectives

Advanced particle filtering

- PFs in high-dimension spaces

- The choice of N in PFs

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Filtering in high-dimension spaces

- All methods, and Monte Carlo is not an exception, suffer from the curse of dimensionality (in this case, when d_x is high)
- In some models, some of the hidden variables can be integrated (Rao-blackwellized PFs):²³
 - implicit dimensionality reduction in the latent space
 - lower variance of PF estimators, compared to working in the original space
- Another approach is to partition the space and run several filters in parallel (Multiple PFs):²⁴
 - implicit dimensionality reduction in the latent space
 - works well when the dimensions of each subset only interact within each subset
 - more research is needed

²³K. Murphy and S. Russell. "Rao-Blackwellised particle filtering for dynamic Bayesian networks". In: *Sequential Monte Carlo methods in practice*. Springer, 2001, pp. 499–515.

²⁴P. M. Djuric, T. Lu, and M. F. Bugallo. "Multiple particle filtering". In: *2007 IEEE International Conference on Acoustics, Speech and Signal Processing-ICASSP'07*. Vol. 3. IEEE. 2007, pp. III–1181.

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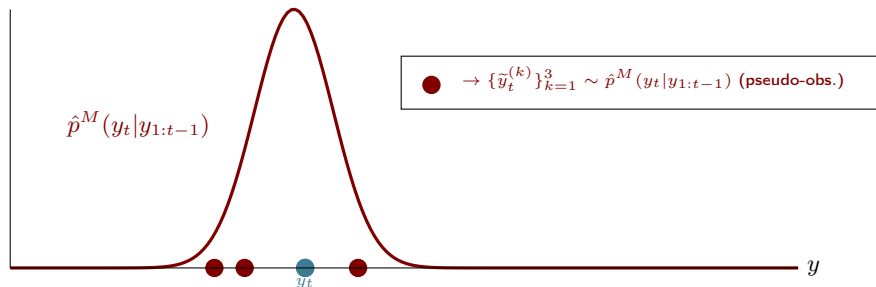
Convergence assessment and adapting N in PF

- **Goal:** in real time and for any SSM:
 1. **evaluate the convergence**, related to the quality of the approximation and
 2. **adapt the number of particles**²⁵,
 3. with theoretical guarantees²⁶
- **Intuition:** check whether the **received observations** “make sense” with the **approximated predictive distributions**
- **Challenge:**
 - at each time step just one observation y_t available
 - the predictive $\hat{p}^M(y_t|y_{1:t-1})$ is evolving with time
- **Proposed method:** At each time step t
 - Generate K fictitious observations $\tilde{y}_t^{(k)}$ from $\hat{p}^M(y_t|y_{1:t-1})$
 1. $\mathbf{x}_t^{(m)} \sim p(\mathbf{x}_t|\tilde{\mathbf{x}}_{t-1}^{(m)})$ (prediction step of BPF, for free)
 2. $\tilde{y}_t^{(k)} \sim \frac{1}{M} \sum_{m=1}^M p(y_t|\mathbf{x}_t^{(m)})$, $k = 1, \dots, K$ (cheap $K \ll M$)
 - Compare them with the actual observation y_t .
 - * Implicitly, we compare $\hat{p}^M(y_t|y_{1:t-1})$ and $p(y_t|y_{1:t-1})$

²⁵V. Elvira, J. Míguez, and P. M. Djurić. “Adapting the number of particles in sequential Monte Carlo methods through an online scheme for convergence assessment”. In: *IEEE Transactions on Signal Processing* 65.7 (2016), pp. 1781–1794.

²⁶V. Elvira, J. Míguez, and P. M. Djurić. “On the performance of particle filters with adaptive number of particles”. In: *Statistics and Computing* 31 (2021), pp. 1–18.

Ordering observation and fictitious observations

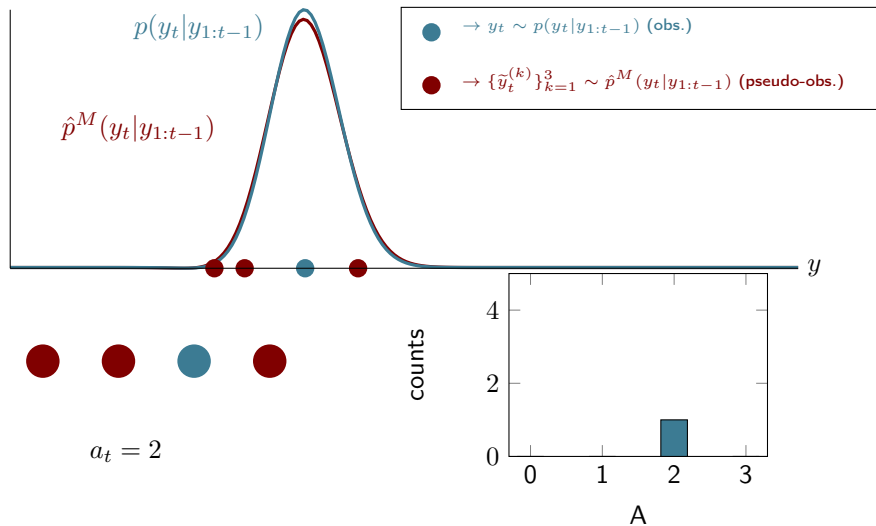


- A_t : number of fictitious observations, $\{\tilde{y}_t^{(k)}\}_{k=1}^3$, smaller than y_t

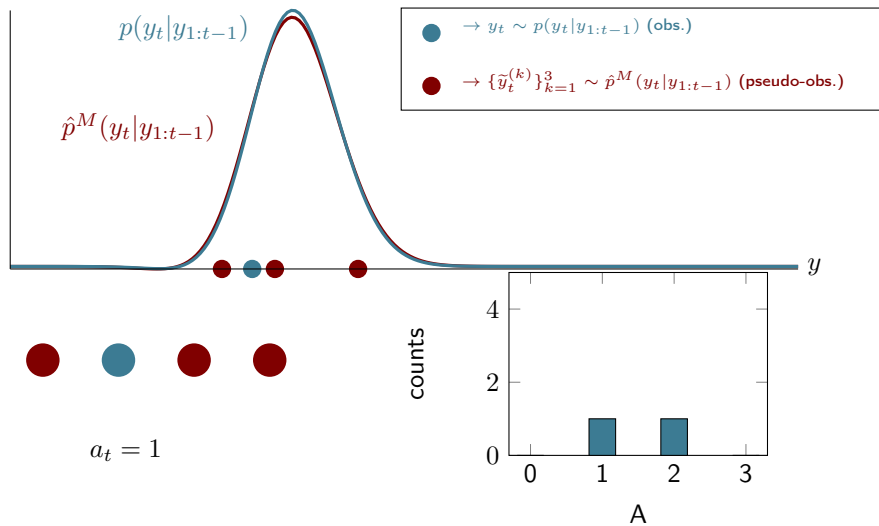


- We can iteratively compute a_t

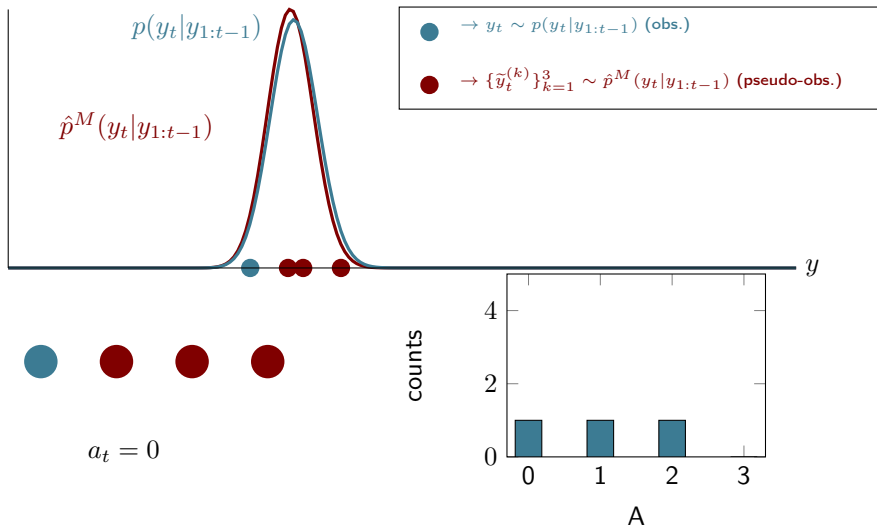
Good approximation, $t = 1$



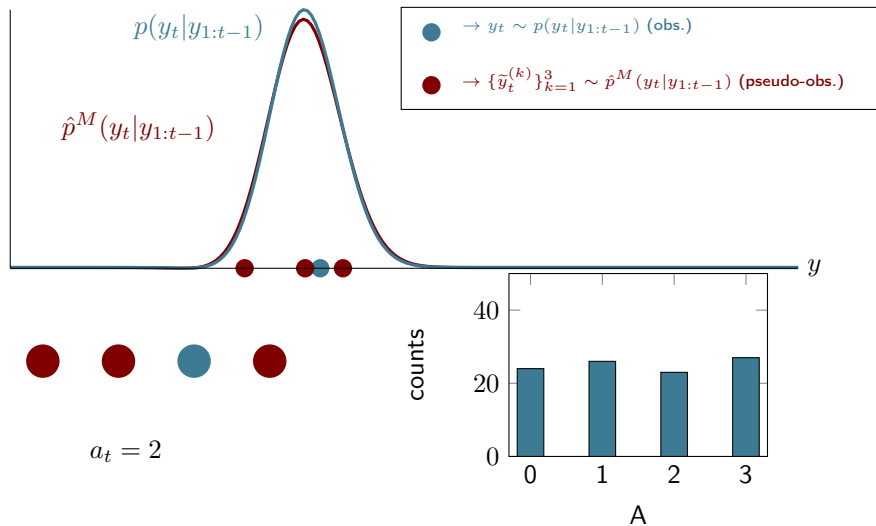
Good approximation, $t = 2$



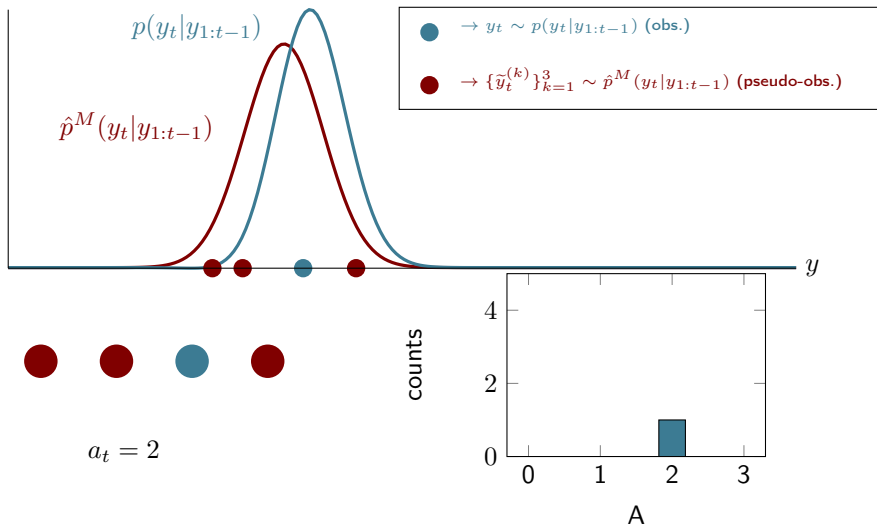
Good approximation, $t = 3$



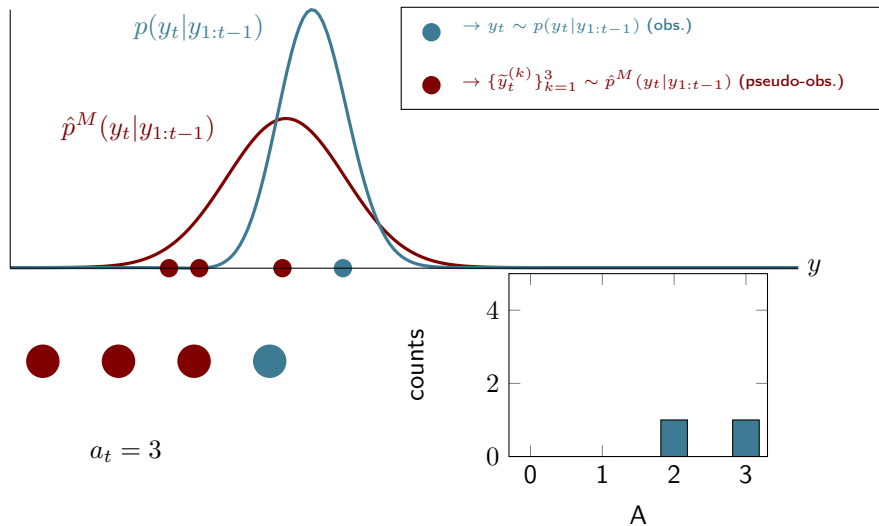
Good approximation, $t = 100$



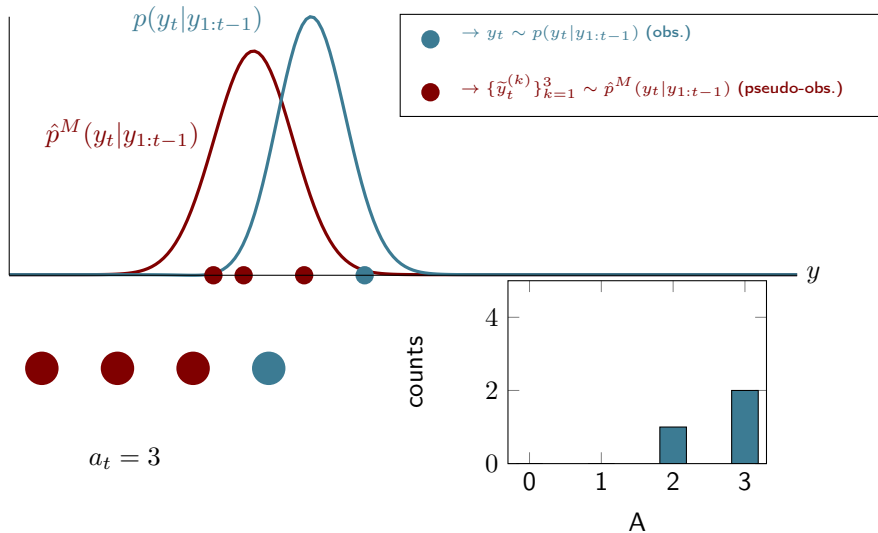
Bad approximation, $t = 1$



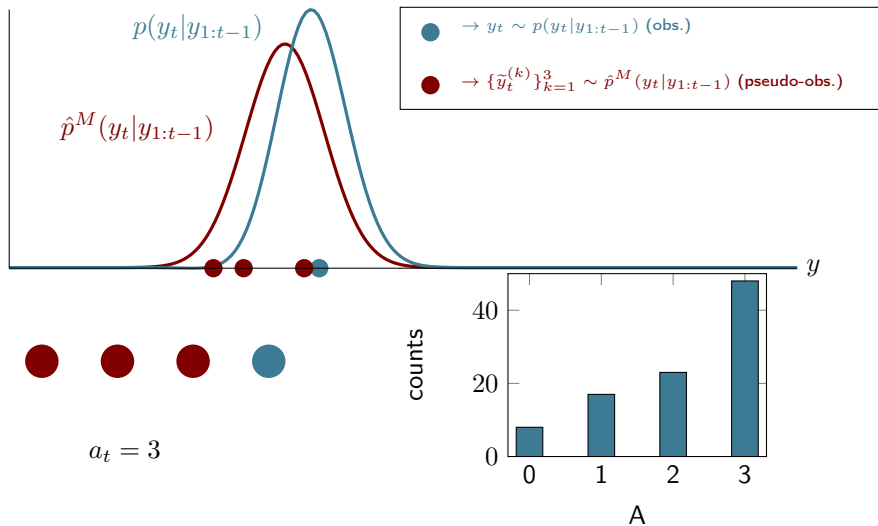
Bad approximation, $t = 2$



Bad approximation, $t = 3$



Bad approximation, $t = 100$



Methodology summary and properties

- **Methodology:** At each time step:
 - simulate $\tilde{y}_t^{(k)} \sim \hat{p}^M(y_t | y_{1:t-1})$, $k = 1, \dots, K$
 - build the r.v. $A_{K,t} := |\mathcal{A}_{K,t}| \in \{0, 1, \dots, K\}$, where $\mathcal{A}_{K,t} := \{y \in \{\tilde{y}_t^{(k)}\}_{k=1}^K : y < y_t\}$
- **Properties:** Under the hypothesis of **perfect approximation**:
 - $\mathcal{J}_t := \{y_t, \tilde{y}_t^{(1)}, \dots, \tilde{y}_t^{(K)}\}$ is a set of i.i.d. samples from a **common** continuous probability distribution $p_t(y_t)$, then:

Proposition 1: *the pmf of the r.v. $A_{K,t}$ is **uniform**:*

$$\mathbb{Q}_K(n) = \frac{1}{K+1}, \quad n = 0, \dots, K.$$

Proposition 2: *the r.v.'s A_{K,t_1} and A_{K,t_2} are **independent**, $\forall t_1, t_2 \in \mathbb{N}$ with $t_1 \neq t_2$.*

- **Invariant wrt the state space model!**

Theoretical results

- **Theoretical analysis:**

- **convergence** of the predictive pdf of the observations:²⁷

$$\lim_{M \rightarrow \infty} \left(f, \hat{p}^M(y_t | y_{1:t-1}) \right) = \left(f, p(y_t | y_{1:t-1}) \right) \quad \text{a.s.,}$$

with explicit **convergence rate**

- ★ extends the existing results of pointwise convergence of $\hat{p}^M(y_t | y_{1:t-1})$ to $\hat{p}(y_t | y_{1:t-1})$
- ★ holds for **multidimensional** observations
- ★ key for the statistical analysis of $A_{K,t}$
- **convergence** of the p.m.f. of $A_{K,t}$ to a discrete uniform distribution

$$\frac{1}{K+1} - \varepsilon_M \leq \mathbb{Q}_K(n) \leq \frac{1}{K+1} + \varepsilon_M, \quad n = 0, \dots, K,$$

with $\lim_{M \rightarrow \infty} \varepsilon_M = 0$ a.s.

- Uniformity of the statistic $A_{K,t}$ (with K fictitious observations) equivalent to $\hat{p}^M(y_t | y_{1:t-1})$ and $\hat{p}(y_t | y_{1:t-1})$ matching K moments.²⁸

²⁷V. Elvira, J. Míguez, and P. M. Djurić. “Adapting the number of particles in sequential Monte Carlo methods through an online scheme for convergence assessment”. In: *IEEE Transactions on Signal Processing* 65.7 (2016), pp. 1781–1794.

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